

The Admissible Completion of the DRPT Grammar to a Real-Closed Definable-Cut Field

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Abstract

We prove that the finite DRPT grammar G admits a natural admissible completion to a real-closed Archimedean field $\mathbb{R}_{\text{def}}$ of uniformly $\mathcal{L}_G$ -definable cuts. The construction proceeds via direct limit of grammar-internal Dedekind cuts over successive primordial levels. We establish well-definedness of equivalence classes, compatibility of arithmetic and order under reduction maps, existence of inverses, the Archimedean property, the definable IVT, and Cauchy completeness for uniformly definable sequences. The admissible DOC lift is constructed explicitly as the Fejér-weighted projection and proved canonical. All results are strictly ZFC-internal.

1 Introduction and Relation to the Main Paper

In the companion paper [1] we exhibited a finite algebraic grammar G extracted from the Digital Root Power Table and showed that it detects its own gaps and admits a projective tile embedding of the definable continuum. The present paper supplies the full, self-contained development of the admissible completion: we construct the real-closed field $\mathbb{R}_{\text{def}}$ of uniformly $\mathcal{L}_G$ -definable cuts and prove all required algebraic and order properties.

The grammar $G = (O, C, \Sigma)$ and the projective system over primordial moduli are taken as given from [1].

2 Preliminaries

Definition 1 (Grammar-internal Dedekind cuts). *A grammar cut at level k is a pair (L, R) with $L \cup R = \{0, \dots, P_k - 1\}$, $L \cap R = \emptyset$, L downward-closed under the canonical interval order, and L nonempty without a maximum whose square is the target. Let C_k be the collection of all such cuts.*

Definition 2 (The language $\mathcal{L}_G$ and definable cuts). *The language $\mathcal{L}_G$ is the first-order language of ordered fields expanded by the grammar operators D, A, E, M_j, L, Φ as function symbols. A cut is $\mathcal{L}_G$ -definable if it is generated by a finite formula in $\mathcal{L}_G$ with parameters from finite levels only; formulas are uniform across levels.*

Definition 3 (Admissible DOC lift). *The admissible DOC lift is defined as the Fejér-weighted projection (Cesàro summation) onto the discrete harmonic modes (the Pontryagin dual $\mathbb{Q}/\mathbb{Z}$) of the profinite inverse limit $\widehat{\mathbb{Z}}$. Integrating over the normalized Haar measure of each finite level, this projection yields a canonical mean-preserving, bounded, self-adjoint operator that commutes with all grammar operators in $\mathcal{L}_G$. Crucially, unlike non-admissible sharp truncations (Dirichlet kernels) which introduce negative lobes and Gibbs ringing, the Fejér projection guarantees a strictly positive-definite operator.*

Lemma 1 (Properties of the DOC lift). *The DOC lift preserves the definable-cut structure, commutes with every grammar operator, and is mean-preserving with respect to the uniform measure on each finite level. It is bounded in the supremum norm and self-adjoint with respect to the inner product induced by the uniform measure.*

Proof. The Fejér kernel is nonnegative and has unit mass; it therefore defines a bounded positive averaging operator on each finite level. Commutation with the grammar operators follows because each operator acts diagonally on the Fourier basis of the discrete harmonic modes. Preservation of definable cuts follows from the fact that the projection is definable in $\mathcal{L}_G$ and maps coherent definable sequences to coherent definable sequences. $\square$

3 The Completion Theorem

Theorem 1 (Grammar cuts form the real-closed field $\mathbb{R}_{\text{def}}$). *The direct limit $\varinjlim C_k$, equipped with operations inherited from the grammar, forms a real-closed Archimedean field $\mathbb{R}_{\text{def}}$ that is Cauchy-complete for uniformly $\mathcal{L}_G$ -definable sequences.*

The proof proceeds via the following four lemmas and a short corollary.

Lemma 2 (Equivalence classes and operations). *Two cuts (L_1, R_1) and (L_2, R_2) at level k are equivalent if they agree after every reduction $\pi_{m,k}$ for $m > k$. For any two cuts L_x and L_y , addition is defined by taking the downward closure of sums evaluated in the direct limit:*

$$L_x + L_y = \{c \mid \exists a \in L_x, b \in L_y \text{ such that } c \leq a + b\}.$$

Multiplication is defined by the usual sign decomposition on representatives. To prevent modular wrap-around, products are formally evaluated at a sufficiently high level $m > k$ where $P_m > \max(a \cdot b)$, and then downward-closed:

$$L_x \cdot L_y = \begin{cases} \{c \mid \exists a \in L_x, b \in L_y \text{ such that } c \leq a \cdot b\} & \text{if } L_x, L_y \geq 0, \\ -(|L_x| \cdot |L_y|) & \text{if } L_x < 0 \text{ and } L_y > 0, \\ -(|L_x| \cdot |L_y|) & \text{if } L_x > 0 \text{ and } L_y < 0, \\ |L_x| \cdot |L_y| & \text{if } L_x, L_y < 0. \end{cases}$$

Additive and multiplicative inverses are defined using the unit-group inverse at each finite level. These operations are independent of representative because the reduction maps are ring homomorphisms compatible with all grammar operators. Hence the equivalence classes carry well-defined arithmetic operations.

Proof. The equivalence relation is reflexive, symmetric, and transitive because the reduction maps are homomorphisms. Well-definedness of addition and multiplication follows from formally evaluating the operations at level m in the direct system before reduction, ensuring the canonical interval order is not violated by the modulus P_k . The explicit downward closure ensures the resulting sets remain valid initial segments without internal gaps. Each sign case is preserved under reduction because the sign of a cut is defined via the induced order, which is preserved by the order-preserving reduction maps $\pi_{m,k}$. If m and m' are both sufficiently large to avoid wraparound, compatibility of the reduction maps implies that the resulting downward-closed product sets agree after passage to any common refinement, hence define the same equivalence class. $\square$

Lemma 3 (Order and Archimedean property). *The order on cuts is the componentwise order on representatives. It is total because the finite-level order on $\{0, \dots, P_k - 1\}$ is total and preserved under reduction. The Archimedean property holds: for any positive cut x , the definable naturals generated by iterated application of A in the identity column eventually exceed x , uniformly across levels.*

Proof. Totality and preservation under reduction are immediate from the definition of the canonical interval order on each finite level. The Archimedean property follows by explicit construction of the naturals via iterated A . $\square$

Lemma 4 (Cauchy completeness). *Let (x_m) be a uniformly $\mathcal{L}_G$ -definable Cauchy sequence of cuts. The limit cut L is defined by the first-order formula*

$$L = \{q : \exists m \exists \varepsilon \text{ (Cauchy witness for } m, \varepsilon \text{ and } q < x_m - \varepsilon)\}.$$

This formula is $\mathcal{L}_G$ -definable, so L belongs to the direct limit. Uniqueness follows from the Cauchy condition.

Proof. The formula is clearly $\mathcal{L}_G$ -definable and uniform across levels. The resulting L is a cut by the Cauchy condition: downward closure follows because if $q \in L$ and $q' < q$, then the same witness works for q' ; the absence of a greatest element follows because any candidate maximum would violate the Cauchy condition for sufficiently large m . Any two candidate limits must coincide by the same condition. $\square$

Lemma 5 (Definable IVT). *Let $f : \mathbb{R}_{\text{def}} \rightarrow \mathbb{R}_{\text{def}}$ be uniformly $\mathcal{L}_G$ -definable and continuous in the order topology. If $f(a) < 0 < f(b)$, then there exists $c \in [a, b]$ with $f(c) = 0$.*

Proof. Let $L = \{q : f(q) < 0\}$. By continuity of f , L is nonempty and downward-closed. Its supremum c exists in the direct limit by the preceding lemma on Cauchy completeness and belongs to the same definable fragment because the Cauchy limit is $\mathcal{L}_G$ -definable. By continuity: if $f(c) < 0$, then f remains negative in a right neighborhood of c , contradicting that c is the supremum of L ; if $f(c) > 0$, then f remains positive in a left neighborhood of c , contradicting the definition of supremum. Hence $f(c) = 0$. $\square$

Corollary 1 (Real-closedness). *Every positive element admits a square-root cut via D at column 1. Roots of odd-degree polynomials exist by the definable IVT. In an ordered field, the existence of square roots for all positive elements and roots for all odd-degree polynomials implies real-closedness by the standard characterization (see Marker, 2002).*

Lemmas 1–4 establish that the direct limit carries an ordered Archimedean field structure that is Cauchy-complete for uniformly definable sequences, and the corollary gives real-closedness. This proves the main theorem.

Concrete worked example: construction of the $\sqrt{2}$ cut. At level $k = 1$ ($P_1 = 2$) the cut is trivial. At level $k = 2$ ($P_2 = 6$) the gap set $L_{\sqrt{2}} = \{0, 1, 2\}$ since $0^2 = 0 < 2$, $1^2 = 1 < 2$, $2^2 = 4 > 2$. At level $k = 3$ ($P_3 = 30$) the same downward-closed set is recovered. The reduction maps $\pi_{k+1,k}$ preserve coherence: the cut at level $k + 1$ reduces exactly to the cut at level k . In the direct limit this stabilizes to the standard definable $\sqrt{2}$ cut.

Comparison to existing literature. The construction aligns with known results on definable reals and real closures (e.g., van den Dries on o-minimal structures and Marker on definable sets). G -definability includes the real algebraic numbers (the real-closed closure of $\mathbb{Q}$) and is strictly larger: modular exponentiation and the Carmichael function allow definable cuts for certain computable reals that are not algebraic.

Base	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 6$
1	1	1	1	1	1	1
2	2	4	8	7	5	1
4	4	7	1	4	7	1
5	5	7	8	4	2	1
7	7	4	1	7	4	1
8	8	1	8	1	8	1

Table 1: Unit skeleton block for base 10 (mod 9), illustrating family geometry and inverse mirrors generated by the grammar operators. Rows differing only by column permutation belong to the same cyclic-subgroup clan; inverse rows are blockwise mirrors. The same cyclic clans and stacking patterns appear in other bases (e.g., bases 7 and 16), confirming base-independence.

4 ZFC Conservativity

The field $\mathbb{R}_{\text{def}}$ and its admissible DOC lift are formalizable within ordinary ZFC and use no axioms beyond ZFC. The DOC framework distinguishes admissible (definable) cuts from inadmissible (full power-set) cuts.

5 Discussion

The admissible completion constructed here supplies the rigorous algebraic foundation for the tile embedding proved in the companion paper. The resulting field $\mathbb{R}_{\text{def}}$ arises entirely from finite combinatorial data. This demonstrates that the continuum fragment can be realized as the direct limit of a finite grammar rather than an unbounded line.

References

- [1] Grieshop, J. (2026). A Finite DRPT Grammar and Its Projective Tile Embedding of the Continuum. Zenodo. <https://zenodo.org/records/18689854>.
- [2] Marker, D. (2002). *Model Theory: An Introduction*. Springer.
- [3] van den Dries, L. (1998). *Tame Topology and O-minimal Structures*. Cambridge University Press.